

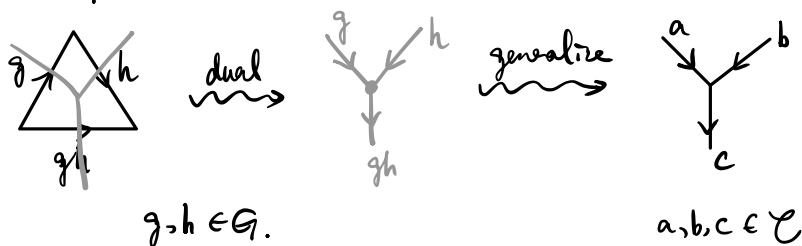
Fusion Categories and Turaev-Viro-Levin-Wen Model

spacetime space

Motivation.

Generalize TQDM of a group G .

Hilbert space:

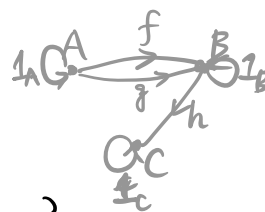


$\mathcal{C}$ $\begin{cases} \text{types: } a, b, c, \dots \\ \text{multiplication / tensor product } a \times b \end{cases}$

3.1. Categories

Def. A category $\mathcal{C}$ consists of

- A collection of objects $\text{Obj}(\mathcal{C}) = \{A, B, \dots\}$
- A collection of morphisms $\text{Hom}(A, B)$ for $\forall A, B \in \text{Obj}(\mathcal{C})$
- A composition map for $\forall A, B, C \in \text{Obj}(\mathcal{C}) = \mathcal{C}$

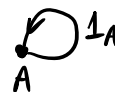


$$C_{A,B,C} : \text{Hom}(A, B) \times \text{Hom}(B, C) \rightarrow \text{Hom}(A, C)$$

$$(f, g) \mapsto g \circ f$$

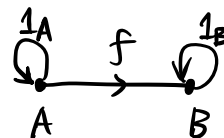
$$\begin{array}{c} \bullet \xrightarrow{f} \bullet \xrightarrow{g} \bullet \\ A \quad B \quad C \end{array} \mapsto \begin{array}{c} \bullet \xrightarrow{g \circ f} \bullet \\ A \quad C \end{array}$$

- An identity morphism $1_A = \text{id}_A \in \text{Hom}(A, A)$



such that

- $(f \circ g) \circ h = f \circ (g \circ h)$ associativity
- $f \circ 1_A = f = 1_B \circ f$ identity



Examples

(1) Set :

$$\text{set } A \xrightarrow{\text{map } f} \text{set } B$$

(2) a group G :



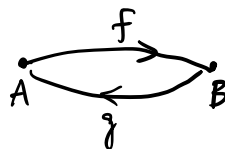
$$\text{Obj}(\mathcal{G}) = \{*\}$$

$$\text{Hom}(*, *) = G$$

$$* \xrightarrow{g} * \xrightarrow{h} * = * \xrightarrow{g \cdot h} *$$

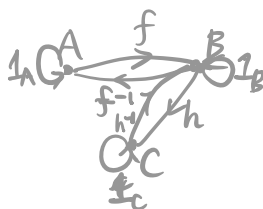
$$1_* = e \in G$$

Def: An isomorphism $f: A \rightarrow B$



$$\begin{cases} g \circ f = 1_A \\ f \circ g = 1_B \end{cases}$$

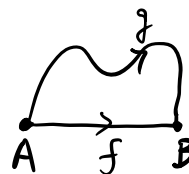
(3) groupoid : A cat $\mathcal{G}$ is a groupoid if every morphism is isomorphism.



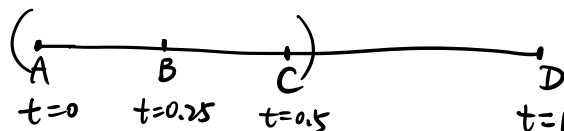
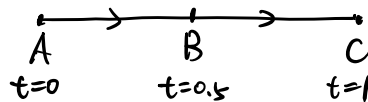
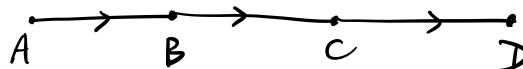
(4) fundamental groupoid of a topological space M :

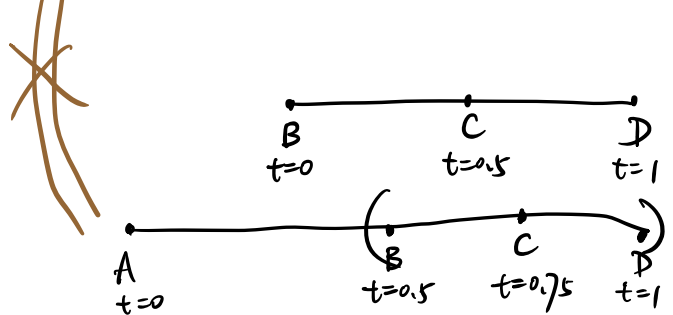
$$\text{Obj} = \{\text{points in } M\}$$

$$\text{Hom}(A, B) = \{\text{Paths from } A \text{ to } B\} / \text{homotopy equivalence}$$



$$\text{path: } I = [0, 1] \rightarrow M$$





$$(5) \text{ Grp : } \text{group } G \xrightarrow[\text{homomorphisms}]{\text{group}} \text{group } G'$$

$$(6) \text{ Vec : } \text{vector space } V \xrightarrow[\text{maps}]{\text{linear}} \text{vector space } W$$

$$(7) \text{ Top : } \text{topological space } M \xrightarrow[\text{maps}]{\text{continuous}} \text{topological space } N$$

Def A functor from categories $\mathcal{C}$ to $\mathcal{D}$ is a map sending

- any object $X \in \mathcal{C}$ to an object Y in $\mathcal{D}$.
- any morphism $f: X \rightarrow Y$ in $\mathcal{C}$ to a morphism $F(f): F(X) \rightarrow F(Y)$ in $\mathcal{D}$.

such that

- F preserves identity : $F(1_X) = 1_{F(X)}$
- F preserves composition : $F(g \circ h) = F(g) \circ F(h)$

$$\begin{array}{ccc} \begin{array}{c} \bullet \xrightarrow{f} \bullet \\ X \quad Y \\ \mathcal{C} \end{array} & \xrightarrow{F} & \begin{array}{c} \bullet \xrightarrow{F(f)} \bullet \\ F(X) \quad F(Y) \\ \mathcal{D} \end{array} \end{array}$$

Examples. (1) $\begin{array}{ccc} * \curvearrowright g \in G & \xrightarrow{F} & V \curvearrowright \rho(g) \\ G & \xrightarrow{F} & \text{Vec} \end{array}$

representation of G .

$$(2) H_n : \text{Top} \longrightarrow \text{Abel}$$

$$\begin{array}{ccc} M & & H_n(M) \\ f \downarrow & \xRightarrow{F} & \downarrow f_* \\ N & & H_n(N) \end{array}$$

Def. Given two functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$, a natural transformation $\alpha: F \Rightarrow G$ assigns to every object X in $\mathcal{C}$ a morphism $\alpha_X: F(X) \rightarrow G(X)$ in $\mathcal{D}$, s.t.

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \downarrow \alpha_X & & \downarrow \alpha_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

$$\begin{array}{ccccc} & & & F(X) & \xrightarrow{F(f)} & F(Y) \\ & & & \downarrow \alpha_X & \searrow \alpha_Y & \\ X & \xrightarrow{f} & Y & \xrightarrow{F} & & \\ & & & \downarrow G & & \\ & & & G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

Example. a group G , given two rep (functors) $\rho: G \rightarrow V$
 $\rho': G \rightarrow V'$

a natural transformation (intertwiner) is a map $f: V \rightarrow V'$,

s.t. $f \circ \rho(g) = \rho'(g) \circ f$ for $g \in G$.

$$\begin{array}{ccccc} & & & V & \xrightarrow{\rho(g)} & V \\ & & & \downarrow f & \searrow f & \\ * & \xrightarrow{g} & * & \xrightarrow{\rho} & & \\ & & & \downarrow \rho' & & \\ & & & V' & \xrightarrow{\rho'(g)} & V' \end{array}$$

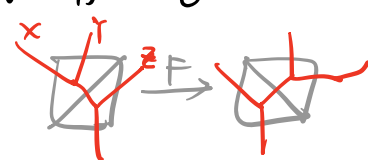
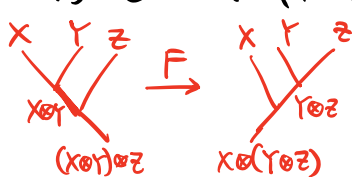
3.2. Fusion categories

Def. A monoidal category consists of

- a category $\mathcal{C}$
- a tensor product functor $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$
- a unit object $1 \in \mathcal{C}$
- a natural isomorphism

$$a_{X,Y,Z}: (X \otimes Y) \otimes Z \rightarrow X \otimes (Y \otimes Z), \quad \forall X, Y, Z \in \mathcal{C}$$

$\uparrow$
associator
F move



- natural isomorphism (left/right unitors) for $X \in \mathcal{C}$

$$l_X: 1 \otimes X \rightarrow X$$

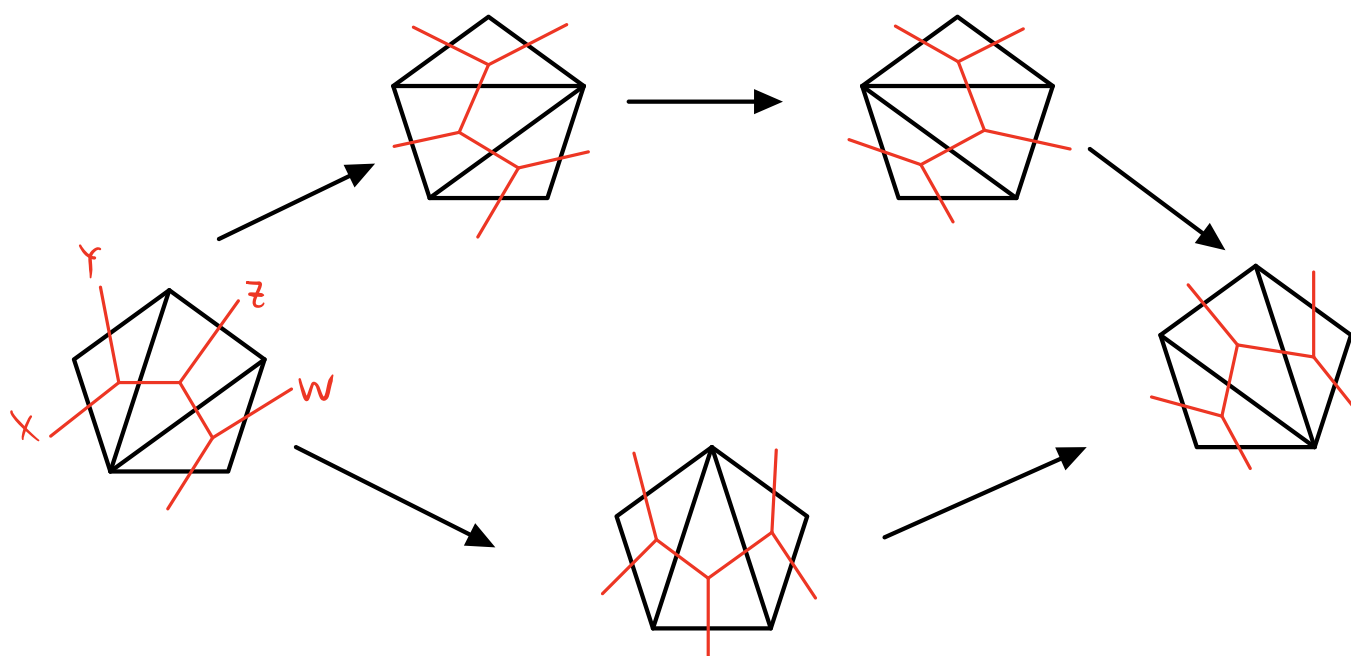
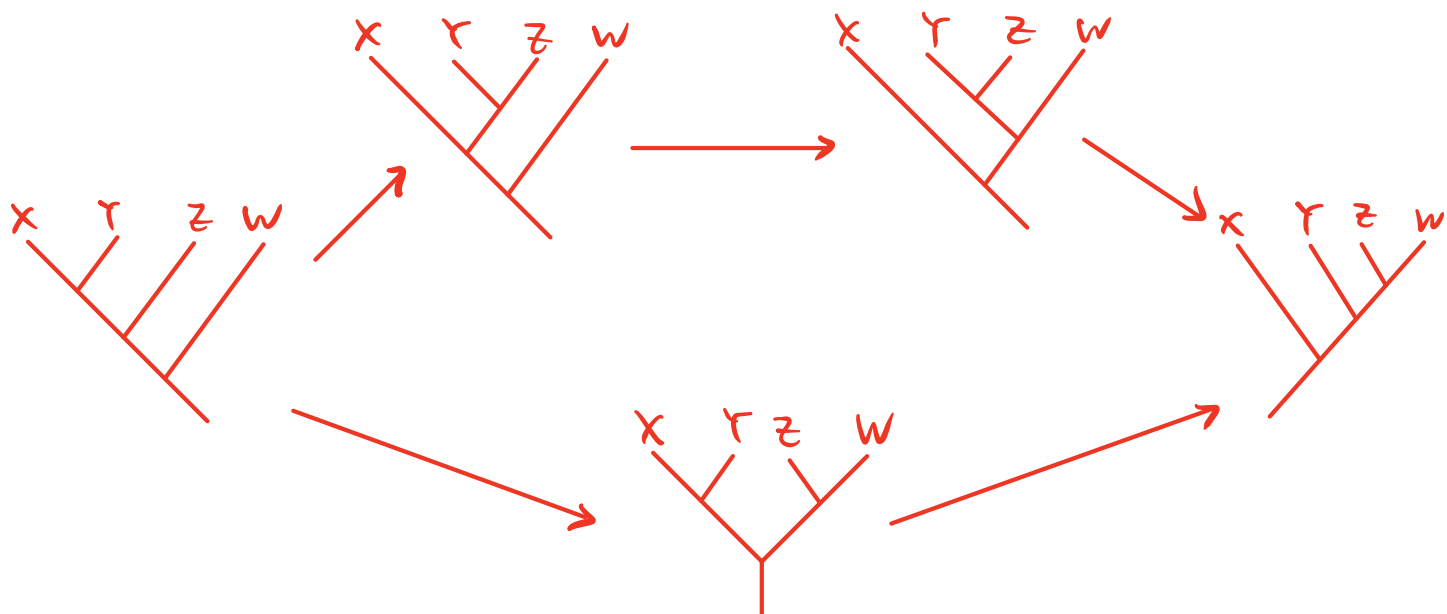
$$r_X: X \otimes 1 \rightarrow X$$

such that

$$\begin{array}{ccc} (X \otimes 1) \otimes Y & \xrightarrow{a_{X,1,Y}} & X \otimes (1 \otimes Y) \\ \downarrow r_X \otimes 1_Y & \searrow \quad \quad \quad \swarrow 1_X \otimes r_Y & \\ & X \otimes Y & \end{array}$$

- pentagon equation:

$$\begin{array}{ccccc} & & & a_{X,Y \otimes Z,W} & \\ & & & \nearrow & \\ & & (X \otimes (Y \otimes Z)) \otimes W & \xrightarrow{a_{X,Y \otimes Z,W}} & X \otimes ((Y \otimes Z) \otimes W) & \xrightarrow{1_X \otimes a_{Y,Z,W}} & X \otimes (Y \otimes (Z \otimes W)) \\ & \nearrow a_{X,Y,Z} \otimes 1_W & & & & \searrow & \\ (X \otimes Y) \otimes (Z \otimes W) & \xrightarrow{a_{X \otimes Y,Z,W}} & (X \otimes Y) \otimes (Z \otimes W) & \xrightarrow{a_{X,Y,Z \otimes W}} & & & \end{array}$$



Examples

(1) $\text{Rep}(G)$.

$\text{obj} = \text{vector space}$

$\text{Hom} = G\text{-invariant linear maps.}$

$$V \otimes W$$

$1 = \text{trivial rep on } \mathbb{C}.$

$\alpha_{x,y,z}$ is trivial

(2) Vec_G .

$\text{obj} = V_g \text{ for } g \in G$

$$\text{Hom}(V_g, V_h) = \begin{cases} \mathbb{C}, & g = h \\ 0, & g \neq h \end{cases}$$

$$V_g \otimes V_h = V_{gh}$$

α_{V_g, V_h, V_k} is trivial

$1 = V_e$, e identity element in G .

(3) $\text{Vec}_G^{\nu_3}$:

$$\alpha_{V_g, V_h, V_k} := \nu_3(g, h, k) \in U(1)$$

$$\text{pentagon eq} \Leftrightarrow d\nu_3 = 1$$

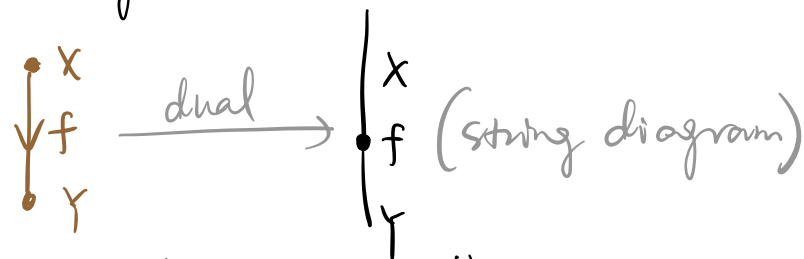
$$\Leftrightarrow \nu_3 \in Z^3(G, U(1))$$

Def.

A fusion category is a rigid, semisimple, linear monoidal category with finitely many isomorphism classes of simple objects such that $\text{Hom}(1, 1) = \mathbb{C}$.

$\begin{array}{ccccc} & \text{with dual} & a \oplus b & \text{hom set is vector space.} & \\ & \uparrow & \uparrow & \uparrow & \\ & \text{rigid} & \text{semisimple} & \text{linear} & \end{array}$

3.3. String diagram

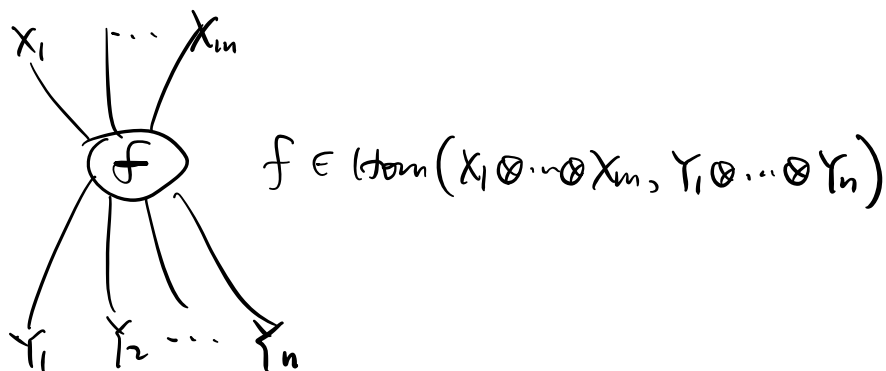
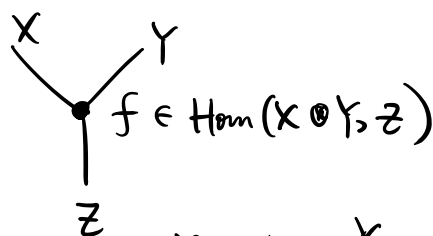


$$\begin{array}{c} X \\ | \\ \bullet \text{ id}_X \\ | \\ X \end{array} = \begin{array}{c} X \\ | \\ X \end{array}, \quad \begin{array}{c} X \\ | \\ \bullet f \\ | \\ Y \\ | \\ \bullet g \\ | \\ Z \end{array} = \begin{array}{c} X \\ | \\ \bullet g \circ f \\ | \\ Z \end{array}$$

tensor: $X \otimes Y$

$$\begin{array}{c} | \\ \otimes \\ | \\ X \quad Y \end{array} \quad \begin{array}{c} X \\ | \\ \bullet f \\ | \\ Y \end{array} \quad \begin{array}{c} X' \\ | \\ \bullet f' \\ | \\ Y' \end{array} = \begin{array}{c} X \otimes X' \\ | \\ \bullet f \otimes f' \\ | \\ Y \otimes Y' \end{array}$$

$$\begin{array}{c} f \\ | \\ \bullet \end{array} \quad \begin{array}{c} | \\ g \\ | \\ \bullet \end{array} = \begin{array}{c} | \\ f \\ | \\ \bullet \end{array} \quad \begin{array}{c} | \\ g \\ | \\ \bullet \end{array} = \begin{array}{c} | \\ f \\ | \\ \bullet \end{array} \quad \begin{array}{c} | \\ g \\ | \\ \bullet \end{array}$$

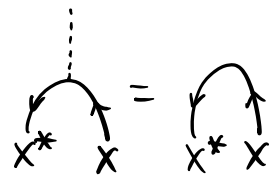


identity 1: $\begin{array}{c} \vdots \\ | \\ \vdots \end{array} =$

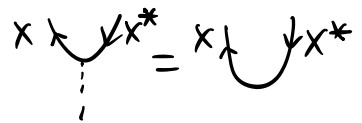
$$\begin{array}{c} X \\ | \\ \bullet f \\ | \\ \vdots \end{array} = \begin{array}{c} X \\ | \\ \bullet f \end{array} \quad f \in \text{Hom}(X, 1)$$

$$\begin{array}{c} \vdots \\ | \\ \bullet f \\ | \\ Y \end{array} = \begin{array}{c} \bullet f \\ | \\ Y \end{array} \quad f \in \text{Hom}(1, Y)$$

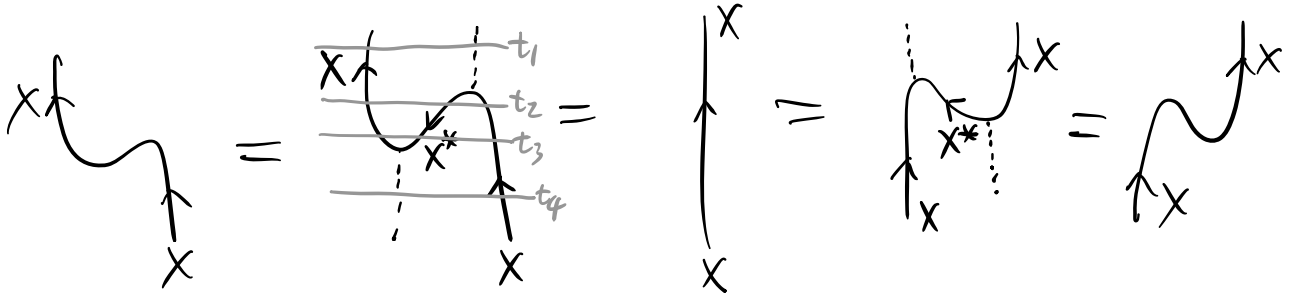
duals: **unit** $i_X: 1 \rightarrow X^* \otimes X$



counit $e_X: X \otimes X^* \rightarrow 1$

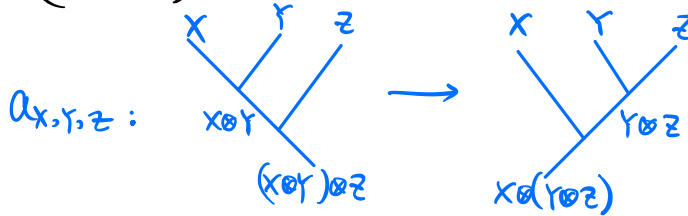


Sit.



$$\begin{array}{ccccccc}
 X \otimes 1 & \xrightarrow{id_X \otimes i_X} & X \otimes (X^* \otimes X) & \xrightarrow{a_{X, X^*, X}^{-1}} & (X \otimes X^*) \otimes X & \xrightarrow{e_X \otimes id_X} & 1 \otimes X \\
 t_1 & & t_2 & & t_3 & & t_4 \\
 & \searrow \gamma_X & & \searrow & & \searrow \ell_X & \\
 & & X & & & &
 \end{array}$$

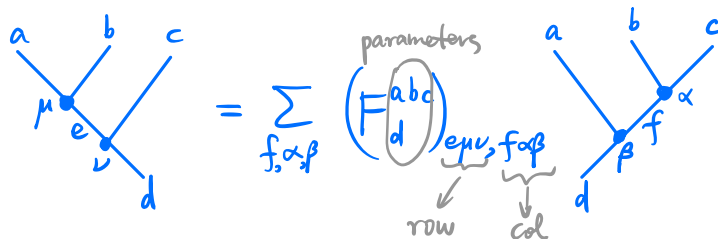
associator (F move):



In simple object basis:

$$V_c^{ab} := \text{Hom}(a \otimes b, c) \ni \begin{array}{c} a \quad b \\ \diagdown \quad \diagup \\ \mu \\ | \\ c \end{array} \leftrightarrow |a, b; c, \mu\rangle$$

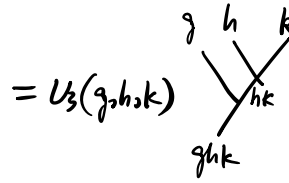
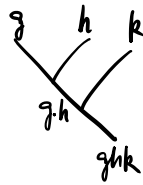
$$N_c^{ab} := \dim V_c^{ab}$$



$$\sum_e N_e^{ab} N_d^{ec} = \sum_f N_f^{bc} N_d^{af}$$

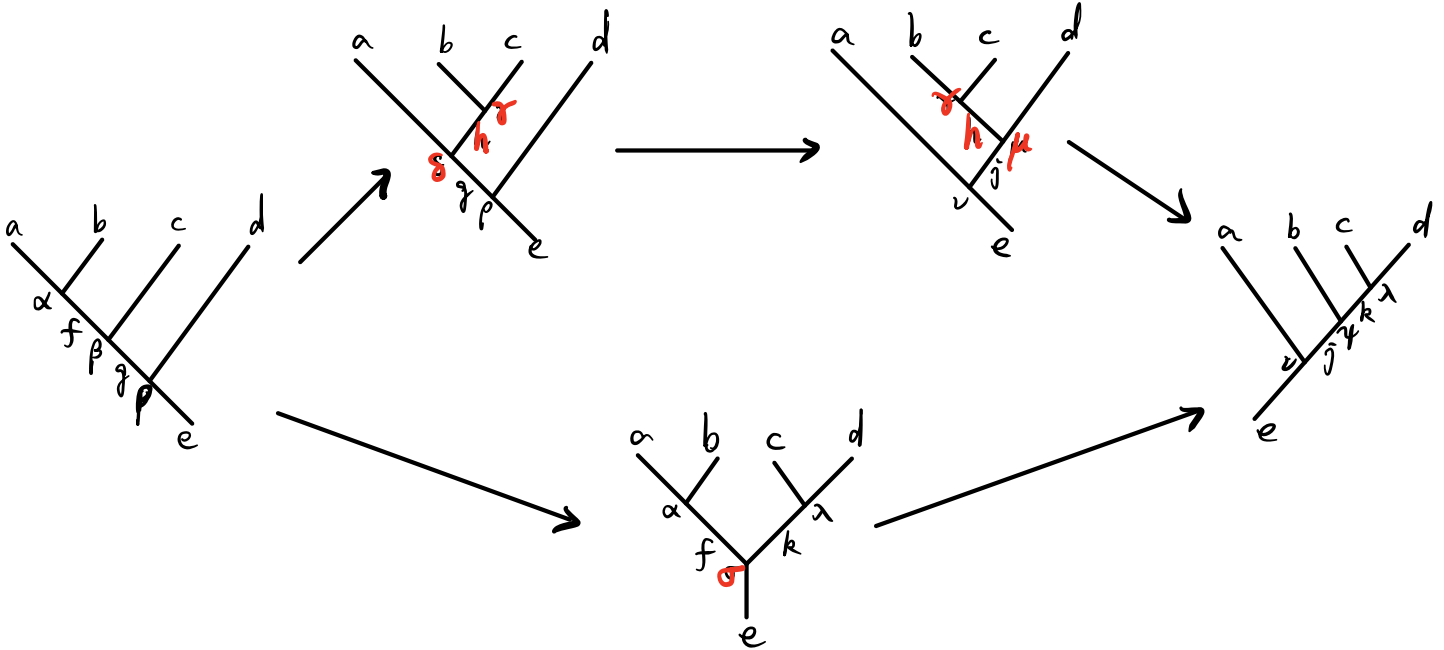
unitary fusion category: F is unitary.

$$\mathcal{C} = \text{Vec}_G^{\mathcal{U}_3} :$$



$$\mathcal{U}_3(g, h, k) := \left(F_{ghk}^{g, h, k} \right)_{1,1}$$

pentagon eq:



$$\sum_{h, \gamma, \delta, \mu} (F_g^{abc})_{f\alpha\beta, h\gamma\delta} (F_e^{abd})_{g\delta\rho, j\mu\nu} (F_j^{bcd})_{h\sigma\mu, k\lambda\tau}$$

$$= \sum_{\sigma} (F_e^{fed})_{g\beta\rho, k\lambda\sigma} (F_e^{abk})_{f\alpha\sigma, j\tau\nu}$$

important relation:

$$\sum_{k, \mu} \begin{array}{c} \text{diagram} \\ \text{diagram} \end{array} = \begin{array}{c} \text{diagram} \\ \text{diagram} \end{array}$$

$$\sum_{k, \mu} |ij; k, \mu\rangle \langle ij; k, \mu| = I$$

$$\begin{array}{c} |k' \\ \text{---} \mu' \\ \bigcirc \text{---} \mu \\ |k \end{array} = \delta_{kk'} \delta_{\mu\mu'} \quad \left| \begin{array}{c} \\ \\ \\ k \end{array} \right.$$

$$\langle i; k' \mu' | i; k \mu \rangle = \delta_{kk'} \delta_{\mu\mu'}$$

quantum dimension:

d_X of a simple object $X \in \mathcal{C}$ is

$$d_X := \bigcirc \nearrow X = X^* \begin{array}{c} 1 \\ \vdots \\ \bigcirc \nearrow X \\ \vdots \\ 1 \end{array} \in \text{Hom}(1, 1) = \mathbb{C}$$

$$d_X > 0.$$

property: $d_i d_j = \sum_k N_k^{ij} d_k$

proof: $d_i d_j = \bigcirc \nearrow i \nearrow j = \sum_{k, \mu} \bigcirc \begin{array}{c} i \mu \\ \bigcirc \nearrow j \\ i \mu \end{array} \nearrow k$

$$= \sum_{k, \mu} \bigcirc \begin{array}{c} j \mu \\ \bigcirc \nearrow i \\ i \mu \end{array} \nearrow k = \sum_k N_k^{ij} \bigcirc \nearrow k$$

$$= \sum_k N_k^{ij} d_k$$

$$d_X = \left[(F_x^{x, \bar{x}, x})_{1,1} \right]^{-1}$$

$$\begin{array}{c} x \quad \bar{x} \quad x \\ \diagdown \quad \diagup \\ \bigcirc \text{---} 1 \\ \diagup \quad \diagdown \\ x \end{array} = (F_x^{x, \bar{x}, x})_{1,1} \begin{array}{c} x \quad \bar{x} \quad x \\ \diagdown \quad \diagup \\ \bigcirc \text{---} 1 \\ \diagup \quad \diagdown \\ x \end{array}$$

$$\bigcirc \nearrow X = (F_x^{x, \bar{x}, x})_{1,1} \bigcirc \nearrow X \bigcirc \nearrow X$$

$$d_X = (F_x^{x, \bar{x}, x})_{1,1} (d_X)^2$$

physical meaning of dx :

$$N_k^{ij} dk = di dj$$

$$(N^i)_k^j dk = di dj$$

$$\cancel{j} \left(\cancel{N^i} \right) \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix} = di \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix}^{\cancel{j}}$$

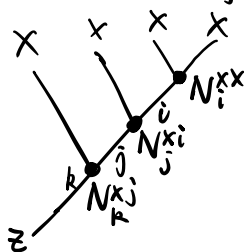
$\Rightarrow (d_1 \dots d_n)^T$ is the eigenvector of N^i with eigenvalue di .

$$N_k^{ij} \geq 0, di > 0$$

$\Rightarrow di$ is the largest eigenvalue of N^i .

Q: what is $\dim$ of $(X)^{\otimes n}$, $n \rightarrow \infty$?

A:



$$\dim X^{\otimes n} := \dim \left[\bigoplus_{\mathbb{Z}} \text{Hom}(X^{\otimes n}, \mathbb{Z}) \right]$$

$$\dim(X)^{\otimes n} = \sum_{i,j,k,\dots} N_i^{XX} N_j^{Xi} N_k^{Xj} \dots$$

$$= \sum_{i,j,k,\dots} (N^X)_i^X (N^X)_j^i (N^X)_k^j \dots$$

$$= \sum_{\mathbb{Z}} \left[(N^X)^n \right]_{X, \mathbb{Z}} \xrightarrow{n \rightarrow \infty} (dx)^n$$

$\Rightarrow \dim(X^{\otimes n}) \sim dx^n$ for $n \rightarrow \infty$

quantum dimension of X .

eg: $\dim \begin{pmatrix} \bullet & \bullet \\ \text{Majorana} & \text{Majorana} \end{pmatrix} = 2$

$$\Rightarrow \dim \begin{pmatrix} \bullet \\ \text{Majorana} \end{pmatrix} = \sqrt{2}$$

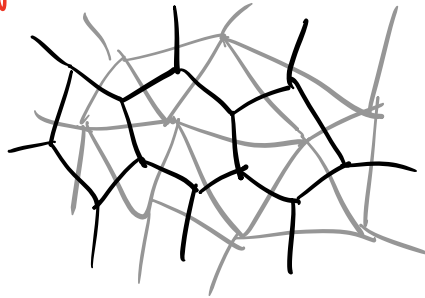
Finally, $\forall$ trivalent graph

$$\text{label} \begin{cases} \text{edge by } X \in \text{Obj}(\mathcal{C}) \\ \text{vertex by } \text{Hom}(X \otimes Y, Z) \text{ or } \text{Hom}(X, Y \otimes Z) \end{cases}$$

$\begin{array}{c} Y \\ \diagup \quad \diagdown \end{array}$
 $\begin{array}{c} \diagup \quad \diagdown \\ Z \end{array}$

$\updownarrow$ dual

triangulation of M_2



3.4. Examples.

Classification of fusion cat ?

Very hard!!! $\mathcal{C} = \text{Vec}_G \rightarrow$ classification of finite simple groups.
(already very hard!)

$$(1) \text{Obj}(\mathcal{C}) = \{1, e\} = \mathbb{Z}_2$$

$\vdots \quad |$

$$e \otimes e = 1, \quad e^* = e$$

associator

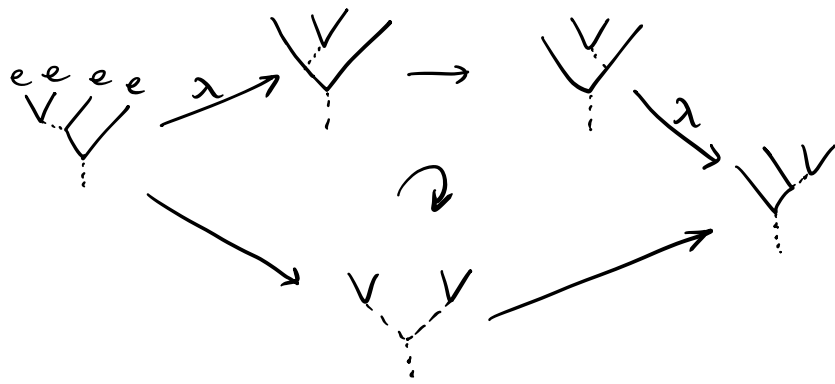
$a_{e,e,e} :$

$$\begin{array}{c} e \quad e \quad e \\ \diagdown \quad \diagup \quad \diagdown \\ \quad \quad \quad e \end{array} = \lambda \begin{array}{c} e \quad e \quad e \\ \diagup \quad \diagdown \quad \diagup \\ \quad \quad \quad e \end{array}$$

$a_{e,e,1} :$

$$\begin{array}{c} \diagdown \quad \diagup \\ \quad \quad \quad \vdots \end{array} = \begin{array}{c} \diagdown \quad \diagup \\ \quad \quad \quad \vdots \end{array}$$

pentagon:



$$\lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

$$H^3(\mathbb{Z}_2, \text{val}) = \mathbb{Z}_2 \ni \nu_3 \begin{cases} \textcircled{1} \nu_3(a,b,c) = 1 \ (\forall a,b,c) : & \lambda = 1 \text{ toric code} \\ \textcircled{2} \nu_3(e,e,e) = -1 & : \lambda = -1 \text{ double semion.} \end{cases}$$

(2) Fibonacci .

$$\bullet \text{Obj}(\mathcal{C}) = \{1, \tau\}$$

$$\bullet \tau \otimes \tau = 1 \oplus \tau \rightsquigarrow \text{non-Abelian}$$

$$N_1^{\tau\tau} = N_\tau^{\tau\tau} = 1$$

$$\tau^{\otimes n} = F_{n-2} 1 \oplus F_{n-1} \tau \quad \text{where } F_n = 1, 1, 2, 3, 5, 8, \dots$$

is the Fibonacci number.



$$\begin{aligned} \tau \otimes (\tau^{\otimes n}) &= \tau \otimes (F_{n-2} 1 \oplus F_{n-1} \tau) = F_{n-2} \tau \oplus F_{n-1} (\tau \otimes \tau) \\ &= F_{n-2} \tau \oplus F_{n-1} (1 \oplus \tau) = F_{n-1} 1 \oplus \underbrace{(F_{n-2} + F_{n-1})}_{F_n} \tau \\ &= F_{n-1} 1 \oplus F_n \tau \end{aligned}$$